MILITARY SPECIFICATION

CONTROLLER, S-2 COMPASS SYSTEM

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers design and all performance requirements for procurement of the S-2 compass system controller.

1.2 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.7.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-5015	Connectors, Electric, "AN" Type
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Plate, Plastic, Lighting

FSC 6605

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:56Z
Check the source to verify that this is the current version before use.

Military (continued)

MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-C-19938	Control, Directional, Electrically Driven Gyro, S-2 Compass and S-5 Automatic Pilot
MIL-A-19939	Amplifier, Two Channel, Integral, S-5 Automatic Pilot, and S-2 Compass
MIL-A-19940	Amplifier, Control, Single Channel, S-2 Compass
MIL-I-21429	Indicator, Compass Repeater, Rotating Dial, Type MW-1
MIL-T-25193	Transmitter, Remote Compass, Type C-2 (Unstabilized)

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Test Levels and Accept/Reject Criteria for Reliability of Non-Expendable Electronic Equipment
MS28079	Controller, S-2 Compass

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality control representative. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:56Z
Check the source to verify that this is the current version before use.

procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 Preproduction - This specification makes provision for preproduction testing.

* 3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(a) Parts and materials requirements shall conform to MIL-E-5400.

(b) Nonrepairable subassemblies, as outlined in MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.5.)

(c) When previously produced models of the controller did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

* 3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-E-5400.

3.3 Design and Construction - The controller shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the controller, excluding cables, shall not exceed 1-3/4 pounds.

3.3.2 Reliability -

* 3.3.2.1 Operational Stability - The controller shall operate with satisfactory performance, continuously or intermittently, for a period of at least 500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:56Z
Check the source to verify that this is the current version before use.

* 3.3.2.2 Operating Life - The controller shall have total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

* 3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The controller shall have 1000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The controller shall provide for the use of cables and connectors in accordance with MIL-E-5400.

* 3.3.3.2 Interconnection Cabling - The controller shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the controller.

3.3.4 Control Panel - The panel of the controller shall conform to the applicable requirements of MIL-C-6781. The configuration of the control panel must be approved by the procuring activity prior to preproduction testing.

3.3.5 Interchangeability of Reordered Equipment - For reordered controllers, interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured controller supplied or designated by the procuring activity.

3.3.6 Interference Control - The generation of radio interference by the controller and the vulnerability of the controller to radio interference shall be controlled within the limits of MIL-I-6181.

* 3.3.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

* 3.3.8 Nameplates - Nameplate approval for controller identification shall be in accordance with MIL-N-18307.

* 3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:56Z
Check the source to verify that this is the current version before use.

Temperature	Room ambient (25 ±5° C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	27.5 ±0.5 V DC

* 3.3.10 Service Conditions - The controller shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment.

* 3.3.11 Primary Input Power Requirements - The controller shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein.

(a) DC Power 28 V, Category "B"

* 3.3.11.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the controller.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.5 Detail Requirements -

3.5.1 Controller - The design of the controller shall provide for selection of both compass controlled directional gyro and free directional gyro operation. The controller shall incorporate means for manually setting the heading of the compass system directional gyro and shall also include a synchronization indicator.

3.5.1.1 Synchronization Indicator - The controller shall include an indicator to show whether the compass system directional gyro is synchronized with the compass system transmitter. The synchronization indicator will

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:56Z
Check the source to verify that this is the current version before use.

further show the direction in which the gyro is being slaved by operation of the set heading-free gyro switch.

3.5.1.2 Synchronization Switch - The controller shall incorporate a three position spring return to center rotary switch designated "SET HEADING - FREE GYRO." The operation of this switch shall subject the compass system directional gyro to fast slaving. Synchronization of the directional gyro to the compass transmitter shall be accomplished by turning the "SET HEADING - FREE GYRO" switch clockwise when the synchronization indicator pointer is indicating to the left of center and turning the switch counterclockwise when the pointer of the synchronization indicator is indicating to the right of center. Turning the rotary switch in "INC" (For Increase) shall slave the repeater indicator(s) in the compass system up-scale and turning the switch to "DEC" (For Decrease) shall slave the repeater indicator(s) down-scale. This manual slaving shall be available only when the system is in "FREE GYRO" operation. In the "SLAVED GYRO" operation, turning the rotary switch shall have no effect on the system.

3.5.1.3 Lighting - The design of the controller panel shall incorporate lighting in accordance with the requirements of MIL-P-7788.

* 3.5.1.4 Case - The outline dimensions of the controller shall conform to MS28079. The case shall be finished with a lusterless black material Color No. 37038 of FED-STD-595. The finishing material shall be of a durable type to withstand usage encountered in service.

3.5.1.5 Electrical Connector - The electrical connector used shall conform to MIL-C-5015 and MS28079. The connector code markings as specified in MIL-E-5400 shall be legibly and permanently applied to the case or connector of the controller.

3.5.1.6 Wiring - The internal wiring shall be as shown in Figure 1. Wiring shall be color coded, neat and accomplished in such a manner that individual wires may be easily traced. Wires shall be secured mechanically to the terminals prior to soldering, where practicable. The wiring shall be insulated from the controller case.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:56Z
Check the source to verify that this is the current version before use.

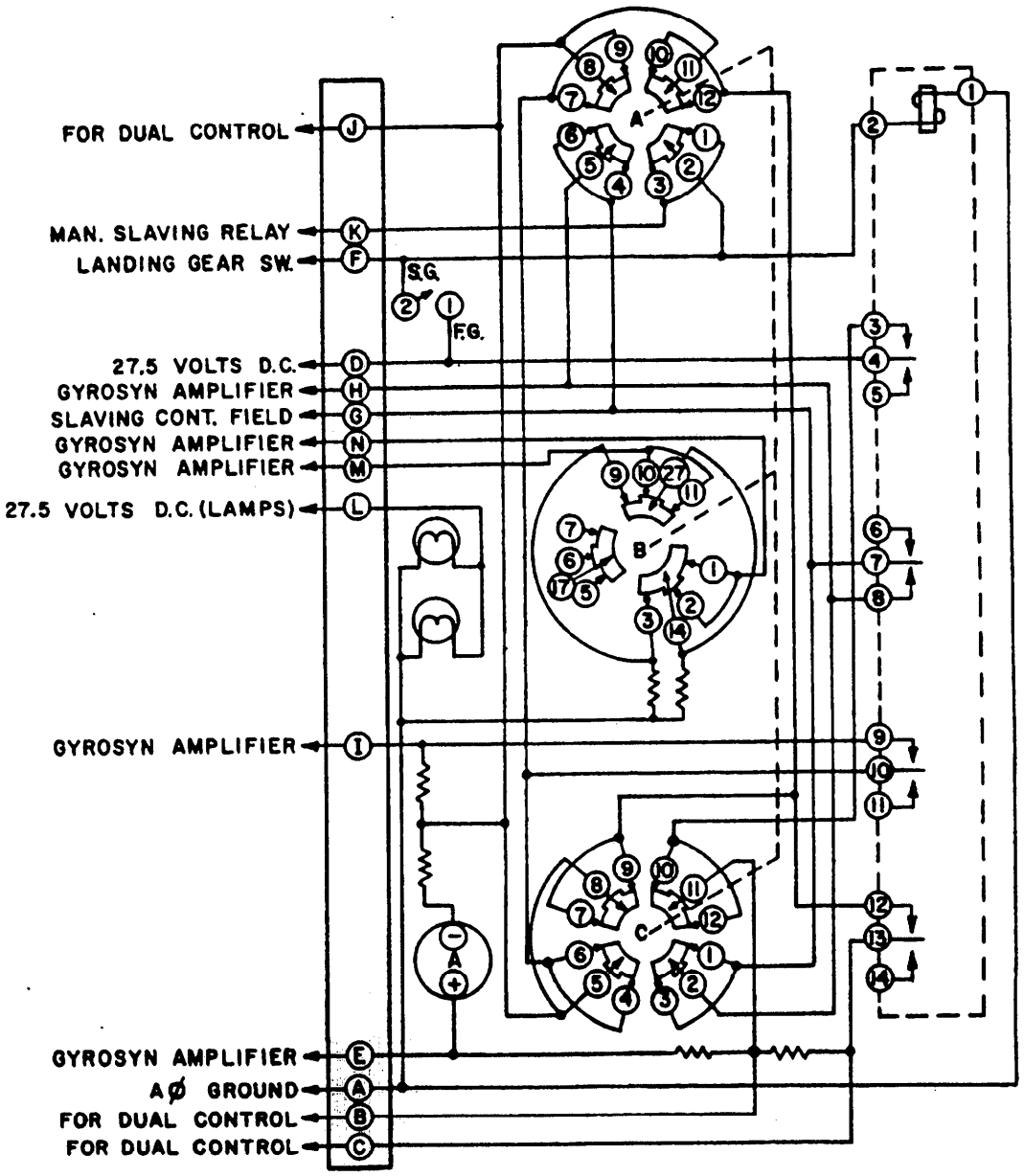


Figure 1. Wiring Diagram

Source: https://assist.dla.mil -- Downloaded: 20261O-19T06:56Z
Check the source to verify that this is the current version before use.

* 4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(a) Preproduction Tests
(b) Initial Production Tests
(c) Acceptance Tests
(d) Longevity Test

* 4.2 Preproduction Tests - Preproduction tests shall be made on a controller representative of the production controllers to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

* 4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

* 4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the controller meets all the requirements of this specification and the contract. As a minimum, preproduction tests shall include all of the tests specified herein under Test Methods.

* 4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production controllers shall be delivered prior to the approval of the preproduction sample. Prefabrication of production controllers prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of controllers to be submitted for acceptance. The preproduction sample shall not be considered as one of the controllers under the contract.

* 4.2.4 Production Equipments - Controllers supplied under the contract shall in all respects, including design, construction, workmanship,

Source: https://assist.dla.mil -- Downloaded: 20260-19T06:56Z
Check the source to verify that this is the current version before use.

performance and quality, be equivalent to the approved preproduction sample. Each controller shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for controllers already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

* 4.3 Initial Production Tests - One of the first ten production controllers shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This controller shall be selected by the procuring activity after the controller has successfully passed all Individual tests. The preproduction sample shall not be selected for this test.

* 4.3.1 Scope of Tests - Each of the controllers selected for the Initial Production Tests shall be subjected to the following tests:

(a)	Sampling Tests	
(b)	Vibration Failure	(4.8.7)
(c)	Radio Noise Interference	(4.8.8)
(d)	High Altitude Cycling	(4.8.9)
(e)	High Temperature Exposure	(4.8.10)
(f)	High Altitude-Low Temperature	(4.8.11)
(g)	Humidity	(4.8.12)
(h)	Fungus Resistance	(4.8.13)
(j)	Salt Spray	(4.8.14)

* 4.3.2 Accessory Material - In addition to the complete controller submitted for Initial Production Tests, the contractor shall also submit such accessory material and data as are necessary to test the controller.

* 4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of controllers under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on controllers previously accepted when requested by the procuring activity.

* 4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test, the controller shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the controller for acceptance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:56Z
Check the source to verify that this is the current version before use.

* 4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual Tests
(b) Sampling Tests
(c) Reliability Assurance Tests
(d) Special Tests

4.4.1 Individual Tests - Each controller submitted for acceptance shall be subjected to the Individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each controller accepted shall have passed the following tests:

(a) Examination of Product	(4.8.1)
(b) Dielectric Strength	(4.8.2)
(c) Continuity	(4.8.3)
(d) Operation	(4.8.4)

* 4.4.2 Sampling Tests - Controllers selected for sampling tests shall first have passed the Individual tests. Controllers shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Controllers Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0 1/
Next 50	1
Next 75	1
Next 100	1
1 for each additional 200 or fraction thereof	

1/ If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one controller from the first 10 produced.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:56Z
Check the source to verify that this is the current version before use.

* 4.4.2.1 Scope of Tests - As a minimum, each controller selected for sampling tests shall be subjected to the following tests:

(a)	Low Temperature Operation	(4.8.5)
(b)	High Temperature Operation	(4.8.6)

* 4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted.

* 4.4.3.1 Qualification Phase - Prior to the acceptance of controllers under the contract or order, a minimum of three (3) controllers shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of controllers to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan I shall be used.

* 4.4.3.2 Reliability Sampling Phase Tests - Samples of the controller shall be tested as outlined in MIL-STD-781, under the section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan II shall be used to determine the length of the tests (until an accept or reject decision is reached).

* 4.4.3.2.1 Lot Size for Sampling Phase - The entire contract quantity (with the exception of those controllers used in the Qualification Phase) shall be considered one lot. The test program shall start the first month after the Qualification Phase has been completed. Samples shall be selected and placed on test in the same manner as though one month's production was a lot. The samples shall be tested until an accept or reject decision is reached. In the meantime, other controllers constructed shall be shipped. The test results shall be summarized monthly for the procuring activity. The procuring activity reserves the right to stop the acceptance of controllers at any time after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the controller, the controller parts, the controller quality control, etc., so that the entire contract quantity will show an accept decision.

* 4.4.3.3 Test Details -The test details such as the length of the Test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests. The following paragraphs shall be considered minimum conditions and shall apply to both the Qualification and Sampling Phases.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:56Z
Check the source to verify that this is the current version before use.

* 4.4.3.3.1 Length of Heat Portion of Cycle - After stabilization at the high temperature limit required by the test level, the controller shall be operated 2-1/2 hours during each cycle and shall be "OFF" for a period of 1/2 hour.

* 4.4.3.3.2 Performance Characteristics to be Measured - The performance characteristics to be measured shall be as specified in 6.8.

* 4.4.3.3.3 Failure Criteria - In addition to the requirements of MIL-STD-781, the following requirements shall be used to determine when a failure has occurred during the tests:

(a) Whenever performance characteristics fall below the acceptance requirement (4.4.3.3.2), at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure unless one caused the other parts to fail.

* 4.4.3.3.4 Preventive Maintenance - During the period of the tests, no preventive maintenance may be performed upon the controller.

* 4.4.4 Special Tests - Special tests shall be conducted on a quantity of controllers for the purpose of checking the effect of any design or material change on the performance of the controllers and to assure adequate quality control. The controllers selected for special tests may be selected from controllers previously subjected to the sampling or reliability assurance tests.

* 4.4.4.1 Special Test Schedule - Selection of controllers for special tests shall be made as follows:

(a) On an early controller after an engineering or material change.
(b) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

* 4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

* 4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(a) Determine the cause of failure.
(b) Determine if the failure is an isolated case or design defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:56Z
Check the source to verify that this is the current version before use.

(c) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.
(d) Where practical, include a test in the individual test to check all controllers for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

* 4.5 Longevity Test - The longevity test shall be performed in accordance with Test Plan XXVIII of MIL-STD-781 except that 5000 hours operating time shall be performed on one sample of the controller. One controller used on the Reliability Qualification Plan may be continued until it has accumulated the required 5000 hours. The longevity test procedures shall be similar to those of 4.4.3. After the first 1000 hours on the longevity test, other maintenance may be performed provided the cost maximum of servicing and replacement of parts throughout the longevity test does not exceed the "cost to repair" specified in the contract. (See 6.2.)

* 4.5.1 Reconditioning of Longevity Test Samples - A controller which has been subjected to the longevity test shall be reconditioned as follows:

(a) On completion of the longevity test, the controller shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.
(b) After reworking, the contractor shall resubmit the controller for acceptance.

* 4.6 Test Procedures - The procedures used for conducting pre-production tests, acceptance tests and longevity tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Test Conditions -

* 4.7.1 Atmospheric Conditions - Unless otherwise specified, all tests required by this specification shall be made in accordance with the requirements of 3.3.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:56Z
Check the source to verify that this is the current version before use.

* 4.7.2 Attitude - Unless otherwise specified, the controller shall be tested in its normal operating position.

* 4.7.3 Lighting - The controller lighting system shall be energized and shall operate satisfactorily during all tests except humidity, fungus resistance, salt spray and as otherwise specified herein. The controller lighting shall operate satisfactorily after completion of each test unless otherwise specified herein.

4.8 Test Methods -

4.8.1 Examination of Product - Each controller shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

4.8.2 Dielectric Strength - A potential of 500 volts (root mean square) alternating current at commercial frequency shall be applied between isolated pins and the case for a period of 5 seconds. There shall be no breakdown of insulation or any other permanent damage to the controller as a result of this test.

4.8.3 Continuity - The continuity of the wiring of each controller shall be carefully examined to determine that the controller is wired in accordance with the requirements of Figure 1.

4.8.4 Operation - This test shall be performed with the controller connected into an operating S-2 compass system.

* 4.8.4.1 With the SLAVED GYRO-FREE GYRO switch in the SLAVED GYRO position, the condition of synchronization of the compass system directional gyro and the repeater indicator(s) headings with the compass transmitter heading shall be shown by the pointer of the synchronization indicator being centered. While the switch is in this position, operation of the SET HEADING FREE GYRO switch shall not cause the directional gyro or the repeater indicator(s) to change heading.

* 4.8.4.2 With the SLAVED GYRO-FREE GYRO switch in the FREE GYRO position, and the system synchronized, operation of the SET HEADING-FREE GYRO switch shall subject the directional gyro of the system to fast slaving. When the SET HEADING-FREE GYRO switch is turned clockwise, the pointer of the synchronization indicator shall move toward the right. When the switch is turned counterclockwise, the pointer of the synchronization indicator shall move toward the left.

* 4.8.4.3 With the SLAVED GYRO-FREE GYRO switch in the FREE GYRO position, operation of the SET HEADING-FREE GYRO switch shall subject

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:56Z
Check the source to verify that this is the current version before use.

the directional gyro of the system to fast slaving. Rotation of the SET HEADING-FREE GYRO switch to the "INC" (increase) position shall cause the directional gyro and the repeater indicator(s) to increase headings. Rotation of the SET HEADING-FREE GYRO switch to the "DEC" (decrease) position shall cause the directional gyro and the repeater indicator(s) to decrease headings.

* 4.8.4.4 Throughout all these tests the two switches and the pointer of the synchronization indicator shall operate smoothly and without binding. Whenever manual operation of the switch is discontinued, the switch shall return to the middle position and the movement of the directional gyro and the repeater indicator(s) shall cease.

* 4.8.5 Low Temperature Operation - The controller shall be properly connected except that no power shall be applied. The controller shall be tested in accordance with Step 2, Temperature and Altitude Tests of MIL-T-5422 for 4 hours. At the end of the 4-hour period and while at the reduced temperature, power shall be applied for 10 minutes. At the end of the 10-minute period and still at the reduced temperature, the controller shall be subjected to and shall meet the requirements of the Operation test.

* 4.8.6 High Temperature Operation - The controller shall be properly connected except that no power shall be applied. The controller shall then be tested in accordance with Step 6, Temperature and Altitude Tests of MIL-T-5422. At the end of the 4-hour period and while still at the elevated temperature, the controller shall be subjected to and shall meet the requirements of the Operation test.

* 4.8.7 Vibration Failure - The controller shall be tested in accordance with Parts I and II, Vibration Tests, of MIL-T-5422. At the conclusion of this test, the controller shall be subjected to and shall meet the requirements of the Individual tests. There shall be no evidence of damage or irregular operation as a result of the vibration.

4.8.8 Radio Noise Interference - The electrical system of the controller shall be tested for conducted and radiated radio noise interference in accordance with MIL-I-6181.

4.8.9 High Altitude Cycling - The controller shall be properly connected into an S-2 compass system with provision made to operate the SLAVED GYRO-FREE GYRO and the SET HEADING-FREE GYRO switches while the controller is at room temperature and a pressure altitude equivalent to 65,000 feet (1.68 in. of Hg. absolute). The SLAVED GYRO-FREE GYRO and the SET HEADING-FREE GYRO switches shall be cycled 500 times each without failure of operation.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T06:56Z
Check the source to verify that this is the current version before use.

4.8.10 High Temperature Exposure - The controller shall be tested in accordance with Step 6, Temperature and Altitude Tests, of MIL-T-5422 for a period of 24 hours. The controller shall be properly connected but no power shall be applied. During the last 15 minutes of the 24-hour period, power shall be applied. At the end of the 24-hour period and while still at the elevated temperature, the controller shall be subjected to and shall meet the requirements of the Operation test. When the controller has been returned to room temperature, the Operation test shall be repeated.

4.8.11 High Altitude-Low Temperature - The controller, properly connected but with no power applied, shall be subjected to Step 3, Temperature and Altitude Tests, of MIL-T-5422 for a period of 72 hours. At the end of the exposure period and while still at the temperature and altitude specified, the controller shall be subjected to and shall meet the requirements of the Operation test. After the controller has returned to room temperature and pressure, the Operation test shall be repeated.

4.8.12 Humidity - The controller shall be subjected to the Humidity Test of MIL-T-5422 except that 71° C shall be substituted for 50° C in Steps 1 and 2. At the conclusion of the test there shall be no evidence of corrosion. The controller shall be subjected to and shall meet the requirements of the Individual tests.

4.8.13 Fungus Resistance - The controller shall be tested in accordance with the Fungus Resistance Test of MIL-T-5422 except that the test period shall be 14 days. At the conclusion of the test, any evidence of the growth of fungi shall be cause for rejection. The controller shall be subjected to and shall meet the requirements of the Individual tests.

4.8.14 Salt Spray - The controller shall be tested in accordance with the Salt Spray Test of MIL-T-5422 except that the test period shall be 50 hours. At the end of the 50-hour period the controller shall be subjected to and shall meet the requirements of the Individual tests. There shall be no evidence of corrosion or other deterioration which could affect subsequent operation.

4.9 Reconditioning of Tested Equipment - Controllers which have been subjected to acceptance and longevity tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the controllers for acceptance.

4.10 Presubmission Testing - No controllers shall be submitted by the contractor until they have been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:56Z
Check the source to verify that this is the current version before use.

4.11 Rejection and Retest - Controllers which have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - Each controller shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Intended Use - The controller covered by this specification is intended for use in conjunction with the S-2 compass system.

* 6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1)
(c) Cost to repair (see 4.5)

* 6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.
(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.
(c) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

* 6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of the controller. Where it appears that a substantial reduction in size and weight or improvement in

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:56Z
Check the source to verify that this is the current version before use.

simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired that their use be investigated. When investigation shows that advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

* 6.5 Non-Repairable Subassemblies - As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

* 6.6 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.7 Associated Equipment - The controller shall operate satisfactorily as part of the S-2 compass system which includes the following components:

COMPONENT	SPECIFICATION
Amplifier, Two Channel, Integral, S-5 Automatic Pilot, and S-2 Compass	MIL-A-19939
Amplifier, Control, Single Channel, S-2 Compass	MIL-A-19940
Control, Directional, Electrically Driven Gyro, S-2 Compass and S-5 Automatic Pilot	MIL-C-19938
Transmitter, Remote Compass, Type C-2 (Unstabilized)	MIL-T-25193
Indicator, Compass Repeater, Rotating Dial, Type MW-1	MIL-I-21429

Source: https://assist.dla.mil -- Downloaded: 202640-19T06:56Z
Check the source to verify that this is the currentversion before use.

6.8 Performance Characteristics to be Measured - At least once each week, the controller shall be subjected to and shall meet the requirements of the Operation Test. (See 4.8.4.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:56Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-C-19941A(AS) CONTROLLER, S-2 COMPASS SYSTEM

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

0-8378

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:56Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Systems Command
Washington, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

NAVAL AIR SYSTEMS COMMAND (AIR-5202l)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T06:56Z
Check the source to verify that this is the current version before use.